LEARNER GUIDE

Adding and subtracting algebraic fractions

Algebraic Expressions and Identities · Lesson 5. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

To add 2/3 and 1/4, rewrite both over 12: 8/12 + 3/12 = 11/12. Algebraic fractions work the same way. Multiply each fraction's numerator and denominator by the same thing — a form of 1, so its value does not change — until the denominators match. Then add or subtract the numerators. When you subtract, the **whole** second numerator is subtracted, so every sign inside it changes. Simplify, and check by putting in a number.

What you are learning to notice

- Only the numerators are added; the common denominator stays: 2/3 + 1/4 is not 3/7.
- With letters: $2/x + 3/(x + 1)$ uses the common denominator $x(x + 1)$.
- Subtracting a bracket: $-2(x - 3) = -2x + 6$.
- If the denominators share a factor, use the lowest common denominator: $3/(x + 1) + 2/(x + 1)^2$ needs only $(x + 1)^2$.
- Check your answer by substituting a value: the sum and your answer must agree.

Moving through it, one step at a time

Find the lowest common denominator. Multiply each numerator and denominator by whatever its denominator is missing. Write one fraction over the common denominator, putting brackets round each numerator. Expand, collect like terms, simplify, and check with a number.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Subtracting a whole numerator

Work out $(x + 1)/2 - (x - 3)/5$.

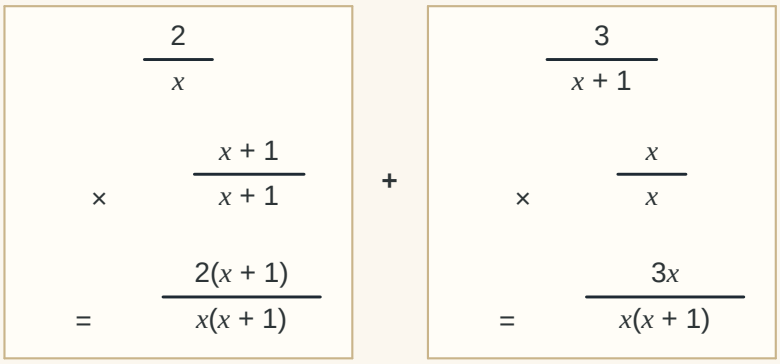
$$\frac{x+1}{2} \times \frac{5}{5} = \frac{5(x+1)}{10} \quad - \quad \frac{x-3}{5} \times \frac{2}{2} = \frac{2(x-3)}{10}$$

1. Both cards now have denominator 10: each was multiplied by a form of 1.	5/5 and 2/2	
2. Combine the numerators, subtracting the whole second one.	$5(x + 1) - 2(x - 3)$	
3. Expand: the –2 changes every sign in $(x - 3)$.	$5x + 5 - 2x + 6$	
4. Collect.	$3x + 11$	

$(x + 1)/2 - (x - 3)/5 = (3x + 11)/10$.

Example 2 — Letters in the denominators

Work out $2/x + 3/(x + 1)$.



1. The common denominator is $x(x + 1)$: each card gets what it is missing.	$\times (x + 1)/(x + 1)$ and $\times x/x$
2. Add the numerators.	$2(x + 1) + 3x = 5x + 2$
3. Check with $x = 1$: $2/1 + 3/2 = 3.5$, and $(5 + 2)/(1 \times 2) = 3.5$.	✓

$2/x + 3/(x + 1) = (5x + 2)/(x(x + 1))$.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (adding number fractions, expanding brackets, and factorising), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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